Diagnosing Causal Claims and Biased Study Designs Answer Key

Precalculus / Advanced Statistics

Quick Answers

1. $\frac{6}{25}$
2. —
3. $\frac{13}{20}$
4. 4
5. 2
6. —
7. 35
8. —

Curriculum

Level: Precalculus / Advanced Statistics
HSS-IC.A–HSS-IC.B – *problems 1, 2, 3, 6, 8*
HSS-ID.A–HSS-ID.C – *problems 4, 5, 7*
Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

3. *If they got 0.24: used the number who answered instead of the number the survey could reach, which measures nonresponse rather than coverage*
4. *If they got 6: dropped the two enrollees who quit before the end and averaged only the finishers*
7. *If they got 2.75: divided the two percentages and reported the ratio as if it were a gap in percentage points*

1. Of the club's 400 members, 96 answered the web survey. What fraction of the membership answered?

Step 1: identify the two counts the question compares — the number who answered (96) over the number who *could* have been counted as members (400). This is the response rate.
Step 2: form the fraction and reduce. Both share a factor of 16: $96 = 16 \cdot 6$ and $400 = 16 \cdot 25$.
Step 3: the reduced fraction is:

$$\boxed{\frac{6}{25}}$$

Step 4: as a decimal that is 0.24, so roughly a quarter of the members answered — and three quarters of the membership is silent in the reported result.

2. Name the biases in a self-selected web survey and say who is over-represented. *(Open response — review by hand.)*
Model answer. Two distinct biases stack here.
Voluntary response bias: the club did not select anyone. It posted a link and let members decide whether to answer, so the sample is self-selected. Volunteers are systematically different from non-volunteers — people with a strong opinion, usually a strong *negative* opinion or an intense interest in the specific proposal, are the ones who take the trouble to respond.
Nonresponse bias: three quarters of the membership did not answer at all, and non-responders are not a random slice of the club. Members who ride occasionally, who joined recently, who do not read the website, or who are simply busy are missing, and their views are missing with them.
Over-represented: the engaged core — frequent riders, long-standing members, officers and their friends, people who visit the club website often, and anyone with a grievance about the specific issue. *Under-represented:* casual, newer, and less-connected members.
Look for: both bias names (voluntary response / self-selection AND nonresponse), plus a concrete description of who answers such surveys. Naming only "small sample" misses the point — the problem is not the size, it is that the sample was assembled by the respondents themselves.

3. Only 260 of the 400 members ever visit the club website. What is the probability a randomly chosen member was reachable by this survey?
Step 1: separate the two ideas the officer merged. *Coverage* asks who *could* be selected — the sampling frame against the population. *Response* asks who answered among those reached. The officer's 0.24 is the response rate from problem 1, so it cannot be the coverage.
Step 2: coverage compares the reachable members (260) with the whole membership (400).
Step 3: reduce 260/400: both share a factor of 20, giving:

$$\boxed{\frac{13}{20}}$$

Step 4: interpret. Coverage is 0.65; roughly a third of the membership was outside the sampling frame entirely and had probability zero of being heard. Undercoverage of that size cannot be repaired by collecting more responses, because the missing members are never reachable — which is exactly why it must not be confused with the response rate.

4. Seven volunteers enrolled; two quit. The company reports a mean loss of 6 kg from the finishers. Compute the mean over everyone who enrolled.
Step 1: decide who belongs in the average. The advertised claim is about what the program does to the people who take it up, so the denominator is everyone who *enrolled*, not everyone who *finished*. Dropping the quitters is attrition bias, and it biases upward whenever the people who leave are the ones doing worst.
Step 2: sum all seven values: $9 + 3 + 12 + 5 + 1 + (-2) + 0 = 28$.
Step 3: divide by seven:

$$\boxed{4}$$

Step 4: compare with the advertised figure. The finishers-only mean is $(9+3+12+5+1)\div 5 = 6$, half again as large. The 6 is arithmetically correct and answers a question nobody asked: "among people for whom the program went well enough that they stayed, how well did it go?"

5. Compute the mean weight change for the seven gym members who did not enroll.

Step 1: the comparison group gets the same treatment in the arithmetic as the program group — all seven values, no exclusions — or the two means are not comparable.
Step 2: sum: $3 + 0 + 5 + 2 + (-1) + 2 + 3 = 14$.
Step 3: divide by seven:

$$\boxed{2}$$

Step 4: line the two up. Program group 4 kg, comparison group 2 kg, an observed difference of 2 kg — and note that the honest comparison uses the all-enrollees mean, not the finishers-only figure the advertisement quoted.

6. Why can these data not support "our program makes you lose twice as much weight"? *(Open response — review by hand.)*

Model answer.
(a) How subjects entered the program group: they **volunteered**. Nobody assigned them. A person who signs up for a twelve-week wellness program is already someone who has decided to change something, has the time and money for it, and expects to succeed. The groups differed before the program began.
(b) A confounding variable: baseline motivation is the cleanest one — volunteers were already trying to lose weight, and would have lost more than the non-volunteers with no program at all. Others that earn full credit: starting weight or BMI, income and the ability to buy the recommended food, existing exercise habits, age, and a doctor's recent instruction to lose weight. The attrition from problem 4 compounds it: the two who quit are exactly the subjects whose data would have pulled the program mean down.
(c) The design that would license the claim: a **randomized controlled experiment**. Recruit a single pool of willing subjects, then assign each one to the program or to a control condition *at random* — a random-number list, not a preference and not the researcher's judgement. Keep everything else parallel between the arms, follow every subject who was randomized (analyze by intention-to-treat, so quitters stay in the arithmetic), and blind the outcome measurement where possible. Random assignment makes the arms alike on average in every variable, measured or not, so a difference in outcome afterwards can be attributed to the program.
Look for: self-selection named explicitly; at least one plausible confounder that would produce the gap on its own; random assignment named as the remedy, not merely "a bigger sample" or "a longer study", neither of which removes the confounding.

7. The sample is 55% retired and the town is 20% retired. The pollster reported the over-representation as 2.75. By how many percentage points are retirees over-represented?

Step 1: name the pollster's error. $55 \div 20 = 2.75$ is a *ratio*: the sample holds 2.75 times the town's share of retirees. "Percentage points" is a subtraction, not a division, and reporting a ratio with the units of a difference makes the bias look smaller than it is.

Step 2: subtract the two shares: $55 - 20$.

Step 3: the over-representation is:

$$\boxed{35}$$

Step 4: read the result as evidence about the design. A weekday-*morning* market systematically over-samples people who are not at work — retirees, shift workers, caregivers — and under-samples the employed. This is a convenience sample, and a gap of 35 percentage points on a single visible variable is a warning that invisible variables are skewed too.

8. Rewrite the headline for what the sample supports, then design the sampling plan that would license it as written. *(Open response — review by hand.)*

Model rewrite. "Among shoppers interviewed at the Saturday-morning farmers' market, a majority opposed the levy." Also acceptable: any conclusion that names the group actually sampled and drops the word "town". What must go: any claim about the town, the electorate, or voters in general.

Model sampling plan. Frame: the county voter-registration list — a list that covers the population the headline is about, unlike "people at a market". *Selection:* a probability sample from that frame — a simple random sample drawn by random number, or a stratified random sample with strata by age and employment status so the known composition of the town is matched, with weights applied afterwards. *Nonresponse:* plan for it before it happens — multiple contact attempts at different times of day and days of the week, several modes (mail, phone, in person), and a follow-up subsample of non-responders to measure how they differ. Report the response rate alongside the estimate.

Look for: a rewritten claim scoped to the people actually sampled; a sampling frame that covers the target population; **random** selection (probability sampling) rather than convenience; and an explicit plan for nonresponse. A plan that just interviews more people at the same market repeats the original bias at a larger sample size, which narrows the confidence interval around the wrong number.